

Vel Tech Group of Institutions

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2028_NeoColab_Design and Analysis of AlgorithmsS1L3

VTU_DAA_Week 3_COD

Attempt : 1

Total Mark : 40

Marks Obtained : 40

Section 1 : Coding

1. Problem Statement

You are given a set of N points on a 2D plane. Your task is to find the distance between the closest pair of points.

Write a program that takes as input the number of points N, followed by the x and y coordinates of each point. The program should output the distance between the closest pair of points, rounded to two decimal places.

Input Format

The first line of input consists of an integer N, representing the number of points.

The following N lines represent the coordinate points x and y, separated by space.

Output Format

The output prints a single floating-point number representing the distance between the closest pair of points, rounded to two decimal places.

Sample Test Case

Input: 3

2 3

12 30

40 50

Output: 28.79

Answer

```
#include <stdio.h>
```

```
#include <math.h>
```

```
int main() {  
    int N;  
    scanf("%d", &N);  
    double x[N], y[N];  
    for(int i=0;i<N;i++) {  
        scanf("%lf %lf", &x[i], &y[i]);  
    }  
    double minDist = 1e9;  
    for(int i=0;i<N;i++) {  
        for(int j=i+1;j<N;j++) {  
            double dx = x[i]-x[j];  
            double dy = y[i]-y[j];  
            double dist = sqrt(dx*dx + dy*dy);  
            if(dist < minDist) minDist = dist;  
        }  
    }  
    printf("%.2f\n", minDist);  
    return 0;  
}
```

Status : Correct

Marks : 10/10

2. Problem Statement

Ishita is working on a mapping application that helps locate the nearest landmarks on a 2D map. Each landmark is represented by its (x, y)

coordinate. Her task is to find out the shortest distance between any two landmarks from the given list.

Help Ishita write a program that calculates the distance between the closest pair of points (landmarks) on the map.

Input Format

The first line of input contains an integer N representing the number of landmarks.

The next N lines contain two space-separated integers, representing the x and y coordinates of each landmark.

Output Format

The output prints a single floating-point number representing the distance between any two landmarks rounded to two decimal places.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 4

5 7

12 32

21 23

43 65

Output: 12.73

Answer

```
#include <stdio.h>
```

```
#include <math.h>
```

```
int main() {
```

```
    int N;
```

```
    scanf("%d", &N);
```

```
    int x[N], y[N];
```

```
    for(int i = 0; i < N; i++) {
```

```
        scanf("%d %d", &x[i], &y[i]);
```

```
    }
```

```

double minDist = 1e9;
for(int i = 0; i < N; i++) {
    for(int j = i + 1; j < N; j++) {
        double dist = sqrt(pow(x[i] - x[j], 2) + pow(y[i] - y[j], 2));
        if(dist < minDist) {
            minDist = dist;
        }
    }
}
printf("%.2f\n", minDist);
return 0;
}

```

Status : Correct

Marks : 10/10

3. Problem Statement

Ragu, an aspiring computer scientist, is currently learning about the concept of finding the closest pair of points in a plane. He wants to test his skills by solving a programming challenge related to this concept.

Given a set of points in a 2D plane, Ragu needs to find the closest pair of points and calculate their Euclidean distance.

Input Format

The first line of input consists of an integer N , representing the number of points in the plane.

The following N lines, each consisting of two space-separated real numbers x and y , represent the coordinates of a point.

Output Format

The first line of output prints the coordinates of the two points forming the closest pair in the below format:

"Closest pair of points: (x_1 , y_1) and (x_2 , y_2)"

The second line prints "Distance: <distance>" representing the Euclidean distance between the closest pair of points, rounded off to two decimal places.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 4

0 3

1 1

10 18

4 9

Output: Closest pair of points: (0, 3) and (1, 1)

Distance: 2.24

Answer

```
#include <stdio.h>
```

```
#include <math.h>
```

```
int main() {
```

```
    int N;
```

```
    scanf("%d", &N);
```

```
    double x[N], y[N];
```

```
    for(int i=0;i<N;i++) scanf("%lf %lf",&x[i],&y[i]);
```

```
    double minDist=1e9;
```

```
    int p1=0,p2=1;
```

```
    for(int i=0;i<N;i++){
```

```
        for(int j=i+1;j<N;j++){
```

```
            double dx=x[i]-x[j];
```

```
            double dy=y[i]-y[j];
```

```
            double dist=sqrt(dx*dx+dy*dy);
```

```
            if(dist<minDist){
```

```
                minDist=dist;
```

```
                p1=i;
```

```
                p2=j;
```

```
            }
```

```
        }
```

```
    }
```

```
    printf("Closest pair of points: (%.0f, %.0f) and (%.0f, %.0f)\n",x[p1],y[p1],x[p2],y[p2]);
```

```
    printf("Distance: %.2f\n",minDist);
```

```
    return 0;
```

```
}
```

Status : Correct

Marks : 10/10

4. Problem Statement

Ragu, a passionate computer science student, is exploring geometry problems to sharpen his algorithmic thinking. Today, he challenges himself to determine the two closest points among a set of 2D coordinates.

Ragu must find the closest pair of points from the given set and compute the Euclidean distance between them with high precision. He also wants the output to display which two points are the closest, formatted clearly.

Input Format

The first line of input contains an integer N representing the number of points in the 2D plane.

The next N lines each contain two space-separated real numbers x and y representing the coordinates of each point.

Output Format

The first line of output prints the coordinates of the two points forming the closest pair in the below format:

"Closest pair of points: (x_1 , y_1) and (x_2 , y_2)"

The second line prints "Distance: <distance>" representing the Euclidean distance between the closest pair of points, rounded off to two decimal places.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5

1 2

3 4

1 0

6 8

2 3

Output: Closest pair of points: (1, 2) and (2, 3)
Distance: 1.41

Answer

```
#include <stdio.h>
#include <math.h>

int main() {
    int N;
    scanf("%d", &N);
    double x[N], y[N];
    for(int i=0;i<N;i++) scanf("%lf %lf",&x[i],&y[i]);
    double minDist=1e9;
    int p1=0,p2=1;
    for(int i=0;i<N;i++){
        for(int j=i+1;j<N;j++){
            double dx=x[i]-x[j];
            double dy=y[i]-y[j];
            double dist=sqrt(dx*dx+dy*dy);
            if(dist<minDist){
                minDist=dist;
                p1=i;
                p2=j;
            }
        }
    }
    printf("Closest pair of points: (%.0f, %.0f) and (%.0f, %.0f)\n",x[p1],y[p1],x[p2],y[p2]);
    printf("Distance: %.2f\n",minDist);
    return 0;
}
```

Status : Correct

Marks : 10/10